

Passivity Properties of the Variable Impedance Controller

To verify the passivity of the proposed adaptive controller, we examine the Variable Impedance Controller's passivity properties, based on the passivity proof proposed in [1]. Let us restate our task-space dynamics

$$M_x(q)\ddot{x} + C_x(q, \dot{q})\dot{x} + G_x(q) = W_c + F_{\text{ext}} \quad (1)$$

where M_x , C_x , and G_x are task-space mass, Coriolis and gravity matrices, respectively, and W_c is the passive impedance controller

$$W_c(x) = G_x(q) - \gamma_d(\cdot)D(x)(\dot{x} - f(x)), \quad \tau_c = J(q)^\top W_c \quad (2)$$

Inserting the controller into the system, we obtain the following closed-loop dynamics

$$M_x(q)\ddot{x} + C_x(q, \dot{q})\dot{x} = -\gamma_d(\cdot)D(x)(\dot{x} - f(x)) + F_{\text{ext}} \quad (3)$$

We can design the damping function $D(x)$ to dissipate energy in orthogonal directions to $f(x)$ formulated as $D(x) = V(x)\Lambda(x)V(x)^\top$:

$$V(x) = [e_1(x) \quad e_2(x)], \quad e_1(x) = \frac{f(x)}{\|f(x)\|}, \quad e_1(x)^\top e_2(x) = 0, \quad \Lambda(x) = \begin{bmatrix} \lambda_1(x) & 0 \\ 0 & \lambda_2(x) \end{bmatrix} \quad (4)$$

as presented in[2], where $\lambda_i(x) \geq 0$ sets the impedance in orthogonal directions of $Q(x)$ basis. This simplifies the controller to:

$$M_x(q)\ddot{x} + (C_x(q, \dot{q}) + \gamma_d(\cdot)D(x))\dot{x} - \gamma_d(\cdot)\lambda_1 f(x) = F_{\text{ext}} \quad (5)$$

The stability and passivity of the robot's closed-loop dynamics can be proven with the following energy storage function:

$$S(x, \dot{x}) = \frac{1}{2}\dot{x}^\top M_x \dot{x} + \gamma_d(\cdot)\lambda_1 \mathcal{V}(x) \quad (6)$$

which includes the robot's kinetic energy and the potential energy injected by the controller term, and accounts for the variable impedance parameter, which also scales the dynamical system in the controller formulation. The potential energy depends solely on the eigenvalue λ_1 and the Lyapunov function $\mathcal{V}(x)$ used to ensure asymptotic stability of $f(x)$, which, for this analysis, we assume to be conservative $f(x) = -\nabla \mathcal{V}(x)$. The time derivative of the energy storage function is,

$$\begin{aligned} \dot{S}(x, \dot{x}) &= \dot{x}^\top M_x \ddot{x} + \frac{1}{2}\dot{x}^\top \dot{M}_x \dot{x} + \gamma_d(\cdot)\lambda_1 \nabla \mathcal{V}^\top \dot{x} + \dot{\gamma}_d(\cdot)\lambda_1 \mathcal{V} \\ &= \dot{x}^\top (F_{\text{ext}} - (C_x + \gamma_d(\cdot)D(x))\dot{x} + \gamma_d(\cdot)\lambda_1 f(x)) + \frac{1}{2}\dot{x}^\top \dot{M}_x \dot{x} + \gamma_d(\cdot)\lambda_1 \nabla \mathcal{V}(x) + \dot{\gamma}_d(\cdot)\lambda_1 \mathcal{V} \\ &= \dot{x}^\top F_{\text{ext}} - \dot{x}^\top \gamma_d(\cdot)D(x)\dot{x} + \frac{1}{2}\dot{x}^\top \underbrace{(\dot{M}_x - 2C_x)}_{=0} \dot{x} + \gamma_d(\cdot)\lambda_1 (f(x) + \nabla \mathcal{V}(x)) + \dot{\gamma}_d(\cdot)\lambda_1 \mathcal{V} \\ &= \dot{x}^\top F_{\text{ext}} - \dot{x}^\top \gamma_d(\cdot)D(x)\dot{x} + \dot{\gamma}_d(\cdot)\lambda_1 \mathcal{V} \end{aligned} \quad (7)$$

which implies that the system will be passive w.r.t the port $(F_{\text{ext}}, \dot{x})$ if

$$\dot{x}^\top F_{\text{ext}} - \dot{x}^\top \gamma_d(\cdot)D(x)\dot{x} + \dot{\gamma}_d(\cdot)\lambda_1 \mathcal{V}(x) \leq \dot{x}^\top F_{\text{ext}} \quad (8)$$

which simplifies to

$$\dot{\gamma}_d(\cdot)\lambda_1 \mathcal{V}(x) \leq \dot{x}^\top \gamma_d(\cdot)D(x)\dot{x}. \quad (9)$$

Since $\lambda_1 \mathcal{V}(x) \geq 0$, this condition can be equivalently written as

$$\dot{\gamma}_d(\cdot) \leq \frac{\dot{x}^\top \gamma_d(\cdot)D(x)\dot{x}}{\lambda_1 \mathcal{V}(x) + \epsilon} \quad (10)$$

where ϵ is a small constant guaranteeing numerical stability. Since we update γ_d recursively, we can approximate its derivative as

$$\dot{\gamma}(\cdot) = \frac{\gamma_{d,k} - \gamma_{d,k-1}}{\Delta t} \quad (11)$$

where Δt is the time difference between parameter computation. Substituting this in the equation results in the following expression

$$\gamma_{d,k} \leq \gamma_{d,k-1} + \Delta t \frac{\dot{x}^\top \gamma_{d,k-1} D(x_{k-1}) \dot{x}}{\lambda_1 \mathcal{V}(x_{k-1}) + \epsilon} \quad (12)$$

which bounds the maximum increase in γ_d over the recursive update. By enforcing this condition on the update, we can guarantee that the system will be passive w.r.t the port $(F_{\text{ext}}, \dot{x})$ when $F_{\text{ext}} \neq 0$, and stable when $F_{\text{ext}} = 0$. Thus, we notice that passivity conditions reduce to

$$\begin{cases} \dot{\gamma}_d \leq 0 & \rightarrow (\text{passive}) \\ \dot{\gamma}_d > 0 \quad \wedge \quad \dot{\gamma}_d(\cdot) \leq \frac{\dot{x}^\top \gamma_d(\cdot) D(x) \dot{x}}{\lambda_1 \mathcal{V}(x) + \epsilon} & \rightarrow (\text{passive}) \\ \dot{\gamma}_d > 0 \quad \wedge \quad \dot{\gamma}_d(\cdot) > \frac{\dot{x}^\top \gamma_d(\cdot) D(x) \dot{x}}{\lambda_1 \mathcal{V}(x) + \epsilon} & \rightarrow (\text{not passive}) \end{cases} \quad (13)$$

meaning that the system will always remain passive when reducing the variable impedance parameter γ_d . This is crucial because the case occurs when human contact is detected, which is when we do not want to break the system's passivity.

While we are always able to enforce the passivity on the system, we noticed that due to the small gradient values, the time that it takes to increase the impedance gains to reach a reactive state after the contact is no longer present increases from about 0.1 seconds to about 76 seconds, which is too slow for most of the systems to maintain high tracking accuracy. Therefore, we utilize the filtered contact estimation, which applies a safety buffer, delaying the increase in impedance gain after the human contact is no longer present. While this solution results in the injection of energy into the system, effectively breaking passivity, it does not happen when a human is in contact with the robot, having no negative impact on the interaction.

We can further examine passivity properties by analyzing the time-derivative of the energy storage function, $\dot{S}(x, \dot{x})$, given as

$$\dot{S}(x, \dot{x}) = \dot{x}^\top F_{\text{ext}} - \dot{x}^\top \gamma_d(\cdot) D(x) \dot{x} + \dot{\gamma}_d(\cdot) \lambda_1 \mathcal{V}. \quad (14)$$

Knowing that passivity is maintained when $\dot{S} \leq \dot{x}^\top F_{\text{ext}}$, we define power balance term of the controller

$$P_{\text{VIC}} = \dot{\gamma}_d(\cdot) \lambda_1 \mathcal{V} - \dot{x}^\top \gamma_d(\cdot) D(x) \dot{x} \quad (15)$$

which implies that the system preserves passivity if $P_{\text{VIC}} \leq 0$. As shown in Figure 1, P_{VIC} exceeds 0 only at the short instant when we increase the variable impedance parameter. However, since this power injection is instantaneous and absent during human-robot interaction, it does not negatively affect the interaction.

References

- [1] Mateusz Jaszczuk and Nadia Figueroa. Rapid mismatch estimation via neural network informed variational inference. In *9th Annual Conference on Robot Learning*, 2025.
- [2] Klas Kronander and Aude Billard. Passive interaction control with dynamical systems. *IEEE Robotics and Automation Letters*, 1(1):106–113, 2015.

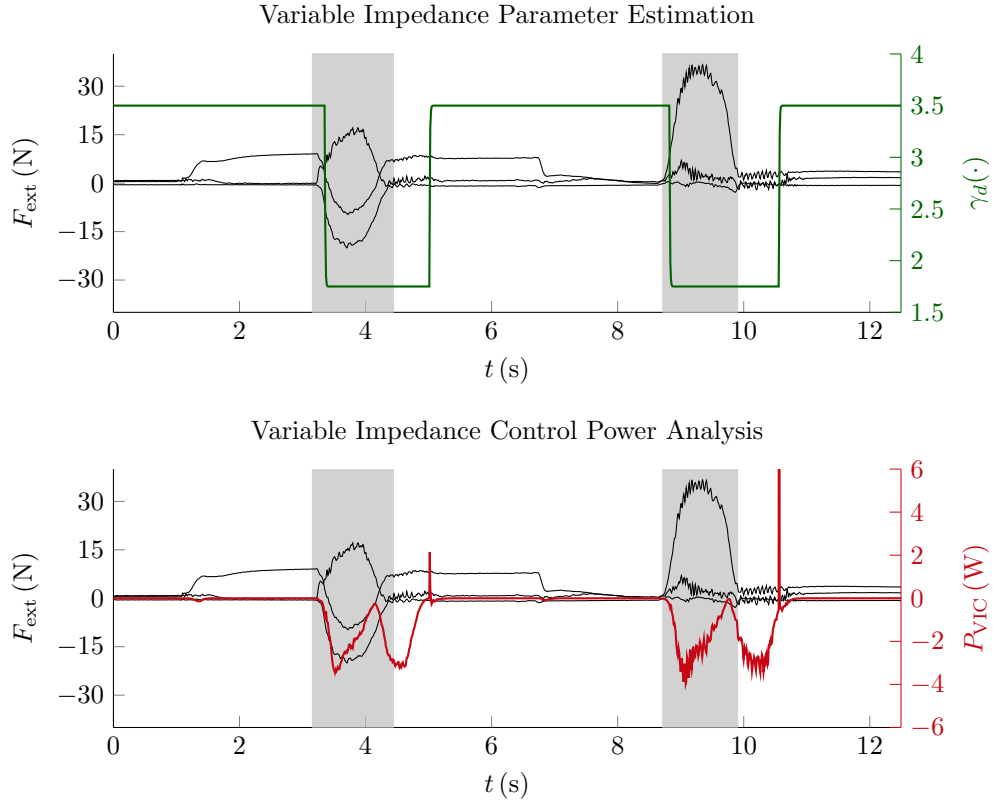


Figure 1: Time history of the variable impedance parameter and P_{VIC} when the adaptive controller detects both loads and perturbations. Gray shaded regions denote human perturbations. The maximum recorded P_{VIC} was observed to be ~ 25 W at the second spike shown in the plot.